專案 1 用SWIFT框架進行LLM微調

https://github.com/robinfang7/nchc_hpc_tutorial/tree/main/swift

swift

LLM training and deployment framework

https://github.com/modelscope/ms-swift

LLaMA-Factory

Easy and Efficient LLM Fine-Tuning

https://github.com/hiyouga/LLaMA-Factory

unsloth

Colossal-AI

步驟說明 (多少量、多少空間)

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Step 1 透過 iService 申請計畫錢包

🤔LLM 大小 ➔ 至少需要多少張 H200, 程式執行時間

🤔用微調訓練的資料集大小 ➔ 需要多少硬碟空間 (預設100 GB /home, /work)

Step 2 使用大家共用的大型計算資源

🚨保護大家資訊安全➔ 雙因子驗證

🚨登入節點禁止執行程式➔ 透過 slurm 使用計算節點

Step 3 建立工作環境與安裝

- **vu venv** or conda or aptainer (or singularity)
- Hugging Face 帳密 (Token)

❓為何選用 vu venv 工作環境➔簡易安裝、軟體更新快、完整

用uv安裝swift (1/2) 放在目錄 swift_uv

[userA@25a-lgn02 ~]$ mkdir swift_uv && cd swift_uv # 新增專案資料夾

[userA@25a-lgn02 ~]$ pip install uv

```
Defaulting to user installation because normal site-packages is not writeable
Collecting uv
  Downloading uv-0.12.5-py3-none-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (23.7 MB)
     |████████████████████████████████| 23.7 MB 2.7 MB/s
Installing collected packages: uv
Successfully installed uv-0.12.5
```

[userA@25a-lgn02 swift_uv]$ uv venv --python 3.12 # 建立python3.12的虛擬工作環境

Using CPython 3.12.13

Creating virtual environment at: .venv

Activate with: source .venv/bin/activate

[userA@25a-lgn02 swift_uv]$ **source .venv/bin/activate** **# 啟用此虛擬工作環境**

❓如何離開此虛擬工作環境 $ deactivate

用uv安裝swift (2/2)

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安裝swift套件

(swift_uv) [userA@25a-lgn02 swift_uv]$ uv pip install ms-swift -U --torch-backend=auto

```
Resolved 151 packages in 6.87s
Prepared 151 packages in 3.81s
[0/151] Installing wheels...
warning: Failed to hardlink files; falling back to full copy. This may lead to degraded performance.
         If the cache and target directories are on different filesystems, hardlinking may not be supported.
         If this is intentional, set `export UV_LINK_MODE=copy` or use `--link-mode=copy` to suppress this wa
Installed 151 packages in 1m 23s
 + absl-py==2.5.0
 + accelerate==1.14.0
 + addict==2.4.0
 + aiofiles==24.1.0
 + aiohappyeyeballs==2.7.1
 + aiohttp==3.14.3
 + aiosignal==1.4.0
 + aliyun-python-sdk-core==2.16.1
 + aliyun-python-sdk-kms==2.16.5
 + annotated-doc==0.0.5
 + annotated-types==0.8.0
 + anyio==4.14.2
 + attrdict==2.0.1
 + attrs==26.1.0
 + binpacking==2.0.1
 + brotli==1.2.0
```

ms-swift==4.5.2
cuda-bindings==13.3.1
trl==0.29.1

下載Huggingface模型與資料集 (1/3)

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套件清單有huggingface-hub，選用access token登入

huggingface-hub==1.28.0

hf-xet==1.6.0

(swift_uv) [userA@25a-lgn05 swift_uv]$ hf auth login

```
? How would you like to log in?  [Use arrows, Enter to confirm]
  Log in with your browser
> Paste an access token
```

V9JD-OBD4

browser

選定後按 author

```
    To log in, `huggingface_hub` requires a token generated from https://huggingface.co/settings/tokens .
Enter your token (input will not be visible):
Token is valid (permission: read).
The token `BLOOM` has been saved to /home/userA/.cache/huggingface/stored_tokens
Your token has been saved to /home/userA/.cache/huggingface/token
Login successful.
The current active token is: `BLOOM`
```

下載Huggingface模型與資料集 (2/3)

(swift_uv) [userA@25a-lgn05 swift_uv]$ hf auth whoami　# 驗證登入者訊息

```
✓ Logged in
  user: ybfang
```

在HFS新增模型與資料的路徑， 下載模型Qwen/Qwen3-4B-Instruct-2507

(swift_uv) [userA@25a-lgn02 swift_uv]$ mkdir -p /work/$(whoami)/llm_models

(swift_uv) [userA@25a-lgn02 swift_uv]$ hf download Qwen/Qwen3-4B-Instruct-2507 --local-dir /work/userA/llm_models/Qwen3-4B-Instruct-2507

```
Fetching 13 files: 100%|                                                    | 13/13 [00:00<00:00, 69.94it/s]
Download complete: :                                          |  0.00B          ✓ Downloadedion complete:
  path: /work/userA/llm_models/Qwen3-4B-Instruct-2507
Download complete: :                                                                          |  0.00B
Reconstruction complete: |                                                              |  0.00B /  0.00B
```

(swift_uv) [userA@25a-lgn02 swift_uv]$ ls /work/userA/llm_models/Qwen3-4B-Instruct-2507

.gitattributes

```
config.json              merges.txt                        model-00003-of-00003.safetensors  tokenizer_config.json
generation_config.json   model-00001-of-00003.safetensors  model.safetensors.index.json      tokenizer.json
LICENSE                  model-00002-of-00003.safetensors  README.md                         vocab.json
```

下載Huggingface模型與資料集 (3/3)

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下載資料集vicgalle/alpaca-gpt4

(swift_uv) [userA@25a-lgn02 swift_uv]$ mkdir -p /work/$(whoami)/data

(swift_uv) [userA@25a-lgn02 swift_uv]$ hf download --repo-type dataset vicgalle/alpaca-gpt4 --local-dir /work/userA/data/vicgalle_alpaca-gpt4

```
Fetching 3 files: 100%|                                                    | 3/3 [00:03<00:00,  1.28s/it]
Download complete: :                                    | 48.2MB, 48.0MB/s  ✓ Downloadedion complete: 100%
  path: /work/userA/data/vicgalle_alpaca-gpt4
Download complete: :                                                       | 48.2MB, 48.0MB/s
Reconstruction complete: 100%|                                           | 48.4MB / 48.4MB, 14.4MB/s
```

(swift_uv) [userA@25a-lgn02 swift_uv]$ ls /work/userA/data/vicgalle_alpaca-gpt4

```
data  README.md
```

train · 52k rows

Search this dataset

instruction string · lengths	input string · lengths	output string · lengths	text string · lengths
9 … 489	0 … 2.47k	1 … 4.57k	166 … 4.93k
Give three tips for staying healthy.		1. Eat a balanced and nutritious diet: Make…	Below is an instruction that describes a task.…
What are the three primary colors?		The three primary colors are red, blue, and…	Below is an instruction that describes a task.…

https://huggingface.co/datasets/vicgalle/alpaca-gpt4

用一張H200對Qwen3-4B模型微調 (1/2)

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編輯腳本run.slurm

```
#!/bin/bash
#SBATCH -A <ProjectID>          # iService Project id
#SBATCH -J swift                # job name
#SBATCH -p dev                  # partition dev
#SBATCH --nodes=1               # Maximum number of nodes to be allocated
#SBATCH --ntasks-per-node=1     # Number of MPI tasks (i.e. processes)
#SBATCH --cpus-per-task=12      # Number of cores per MPI task
#SBATCH --gres=gpu:1
#SBATCH -o %x_%j.out            # Path to the standard output file
```

source .venv/bin/activate

用一張H200對Qwen3-4B模型微調 (2/2)

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編輯腳本run.slurm (cont.)

```
CUDA_VISIBLE_DEVICES=0 \
swift sft \
    --model /work/userA/llm_models/Qwen3-4B-Instruct-2507 \
    --tuner_type lora \
    --dataset /work/userA/data/vicgalle_alpaca-gpt4 \
    --torch_dtype bfloat16 \
    --num_train_epochs 1 \
    --per_device_train_batch_size 1 \
    --per_device_eval_batch_size 1 \
    --learning_rate 1e-4 \
    --lora_rank 8 \
    --lora_alpha 32 \
    --target_modules all-linear \
    --gradient_accumulation_steps 16 \
```

```
--eval_steps 50 \
--save_steps 50 \
--save_total_limit 2 \
--logging_steps 5 \
--max_length 2048 \
--output_dir output \
--warmup_ratio 0.05 \
--dataloader_num_workers 4 \
--model_author swift \
--model_name swift-robot
```

提交作業

透過 slurm 在計算節點執行

[userA@25a-lgn05 swift_uv]$ sbatch run.slurm

```
Submitted batch job 287023
```

[userA@25a-lgn05 swift_uv]$ squeue -u userA

```
 JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
287023       dev    swift userA  R       4:12      1 25a-hgpn001
```

[userA@25a-lgn05 swift_uv]$ ssh 25a-hgpn001

```
Last login: Wed Aug  5 15:46:05 2026 from 172.21.x.x
```

[userA@25a-hgpn001 ~]$ watch nvidia-smi

```
Every 2.0s: nvidia-smi                                  25a-hgpn001: Fri Aug 21 14:00:46 2026

Fri Aug 21 14:00:46 2026
+-----------------------------------------------------------------------------------------+
| NVIDIA-SMI 580.65.06              Driver Version: 580.65.06      CUDA Version: 13.0     |
+-----------------------------------------+------------------------+----------------------+
| GPU  Name                 Persistence-M | Bus-Id          Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |           Memory-Usage | GPU-Util  Compute M. |
|                                         |                        |               MIG M. |
|=========================================+========================+======================|
|   0  NVIDIA H200                    On  |   00000000:DA:00.0 Off |                    0 |
| N/A   59C    P0            607W /  700W |  127253MiB / 143771MiB |     100%      Default |
|                                         |                        |             Disabled |
+-----------------------------------------+------------------------+----------------------+

+-----------------------------------------------------------------------------------------+
| Processes:                                                                              |
|  GPU   GI   CI              PID   Type   Process name                        GPU Memory |
|        ID   ID                                                               Usage      |
|=========================================================================================|
|    0   N/A  N/A         1082305      C   ...16/swift_uv/.venv/bin/python3      12724... |
+-----------------------------------------------------------------------------------------+
```

Benchmark (1/3)

Batch size = 1

```
Train:   1%|          | 17/3251 [01:38<4:39:22,  5.18s/it]
```

```
+-----------------------------------------------------------------------------------------+
| NVIDIA-SMI 580.65.06              Driver Version: 580.65.06      CUDA Version: 13.0     |
+-----------------------------------------+------------------------+----------------------+
| GPU  Name                 Persistence-M | Bus-Id          Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |           Memory-Usage | GPU-Util  Compute M. |
|                                         |                        |               MIG M. |
|=========================================+========================+======================|
|   0  NVIDIA H200                    On  |   00000000:9A:00.0 Off |                    0 |
| N/A   40C    P0            158W /  700W |   13273MiB / 143771MiB |      6%      Default |
|                                         |                        |             Disabled |
+-----------------------------------------+------------------------+----------------------+


+-----------------------------------------------------------------------------------------+
| Processes:                                                                              |
|  GPU   GI   CI              PID   Type   Process name                        GPU Memory |
|        ID   ID                                                               Usage      |
|=========================================================================================|
|    0   N/A  N/A         2655200      C   ...e2/swift_uv/.venv/bin/python3      13264MiB |
+-----------------------------------------------------------------------------------------+
```

Benchmark (2/3)

Batch size = 16

```
Train:   1%|          | 3/204 [00:43<48:18, 14.42s/it]
```

```
+-----------------------------------------------------------------------------+
| NVIDIA-SMI 580.65.06              Driver Version: 580.65.06      CUDA Version: 13.0     |
+-----------------------------------------+------------------------+----------------------+
| GPU  Name                 Persistence-M | Bus-Id          Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |           Memory-Usage | GPU-Util  Compute M. |
|                                         |                        |               MIG M. |
|=========================================+========================+======================|
|   0  NVIDIA H200                    On  |   00000000:18:00.0 Off |                    0 |
| N/A   57C    P0            590W /  700W | 109697MiB / 143771MiB  |     100%      Default |
|                                         |                        |             Disabled |
+-----------------------------------------+------------------------+----------------------+


+-----------------------------------------------------------------------------+
| Processes:                                                                  |
|  GPU   GI   CI              PID   Type   Process name                  GPU Memory |
|        ID   ID                                                          Usage      |
|=============================================================================|
|    0   N/A  N/A         1667385      C   ...e2/swift_uv/.venv/bin/python3      10968... |
+-----------------------------------------------------------------------------+
```

Benchmark (3/3)

Batch size =64

```
Train:   0%|          | 0/51 [00:00<?, ?it/s][INFO:swift] use_logits_to_keep: True
[W824 11:54:15.886866724 CUDACachingAllocator.cpp:3933] memory allocation failed with OOM on device 0
[W824 11:54:19.996719207 CUDACachingAllocator.cpp:3933] memory allocation failed with OOM on device 0
Train:   2%|          | 1/51 [00:58<48:26, 58.14s/it]
```

```
+-----------------------------------------------------------------------------+
| NVIDIA-SMI 580.65.06              Driver Version: 580.65.06      CUDA Version: 13.0     |
+-----------------------------------------+------------------------+----------------------+
| GPU  Name                 Persistence-M | Bus-Id          Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |           Memory-Usage | GPU-Util  Compute M. |
|                                         |                        |               MIG M. |
|=========================================+========================+======================|
|   0  NVIDIA H200                    On  |   00000000:18:00.0 Off |                    0 |
| N/A   59C    P0            677W /  700W | 137975MiB / 143771MiB |     100%      Default |
|                                         |                        |             Disabled |
+-----------------------------------------+------------------------+----------------------+

+-----------------------------------------------------------------------------+
| Processes:                                                                  |
|  GPU   GI   CI              PID   Type   Process name                  GPU Memory |
|        ID   ID                                                          Usage      |
|=============================================================================|
|    0   N/A  N/A         1669746      C   ...e2/swift_uv/.venv/bin/python3      13796... |
+-----------------------------------------------------------------------------+
```

常用資訊

https://man.twcc.ai/@nano4-manual/documentation

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thank you
danke
謝謝
ngiyabonga
teşekkür ederim
gracias
merci
obrigado
bedankt
dziękuję
go raibh maith agat
mochchakkeram
sukriya
kop khun krap
terima kasih
grazie
arigatô
dakujem
takk
tapadh leat
спасибо
рахмат
Баярлалаа
mersi
barka
welalin
tack
spas
dank je
misaotra
matondo
paldies
grazzi
mahalo
vinaka
спасибі
blagodaram
faafetai lava
kiitos
dankie
dhanyavad
nanni
nandri
bayarlalaa
gràcie
hvala
mauruuru
köszönöm
akun
dankon
ačiū
хвала
asante
manana
obrigada
tenki
murakoze
chokrane
djiere dieuf
tau
дякую
mamnun
enkosi
chnorakaloutioun
gratias ago
gràcies
sulpáy
taiku
sobodi
dēkuji
sagolun
mèsi
didi madloba
najis tuke
kam sah hamnida
rahmat
ありがとう
tanemirt
rahmet
diolch
dhanyavadagalu
shukriya
merce
мерси
trugarez
তোমাকে ধন্যবাদ
감사합니다
xièxie
ευχαριστώ